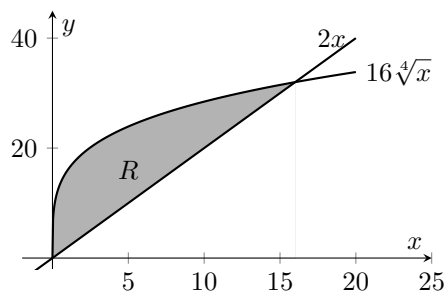


The disk/washer method

Let R be the region bounded by the following curves. Use the washer or disk method to find the volume generated when R is revolved about the x -axis.

24. from 6.3 $y = 16\sqrt[4]{x}$ and $y = 2x$.



The intersections points:

$$\begin{aligned} 16x^{1/4} &= 2x \\ (2^4 x^{1/4})^4 &= (2x)^4 \\ 2^{16} x &= 2^4 x^4 \\ 2^4 x(2^{12} - x^3) &= 0 \Rightarrow x = 0 \text{ and } x = 16 \end{aligned}$$

The general formula is:

$$\text{Volume} = \pi \int_a^b \underbrace{f(x)^2}_{\text{top}} - \underbrace{g(x)^2}_{\text{bottom}} dx \quad (\text{NOT: } \int_a^b (f(x) - g(x))^2 dx)$$

$$\text{So we have, Volume} = \pi \int_0^{16} (16x^{1/4})^2 - (2x)^2 dx = \pi \left(\frac{512}{3} x^{3/2} - \frac{4x^3}{3} \right) \Big|_0^{16} = \frac{16384}{3} \pi \text{ cubic units.}$$